

Logic Symbols & Greek Letters Cheat Sheet

Logic Symbols Cheat Sheet (with LaTeX)

Symbol	LaTeX Code	Name	Pronunciation	Meaning / Usage
$\models$	<code>\models</code>	Double turnstile	models / satisfies	Semantic satisfaction (state or system satisfies a formula)
$\not\models$	<code>\not\models</code>	Not models	does not satisfy	The formula does not hold in the model
$\vdash$	<code>\vdash</code>	Single turnstile	proves	Syntactic derivation (provable from assumptions)
$\neg$	<code>\neg</code>	Negation	not	Logical NOT
$\wedge$	<code>\wedge</code>	Conjunction	and	Logical AND
$\vee$	<code>\vee</code>	Disjunction	or	Logical OR
$\rightarrow$	<code>\rightarrow</code>	Implication	implies	If... then...
$\leftrightarrow$	<code>\leftrightarrow</code>	Biconditional	iff	If and only if (logical equivalence)
$\forall$	<code>\forall</code>	Universal quantifier	for all	Property holds for all paths or objects
$\exists$	<code>\exists</code>	Existential quantifier	there exists	Property holds for at least one path or object
$\diamond / \blacklozenge$	<code>\Diamond /</code>	Eventually	eventually	Holds at some

	<code>\lozenge</code>			point in the future
□ / ◻	<code>\Box / \square</code>	Always	always / globally	Holds at all times in the future
◯	<code>\bigcirc</code>	Next	next	Holds in the next state
U	U	Until	until	One condition holds until another becomes true
R	R	Release	release	Dual of 'until'
∈	<code>\in</code>	Element of	is in	Membership in a set
⊆	<code>\subseteq</code>	Subset	is a subset of	One set is contained in another
→	<code>\to</code>	Transition	transitions to	Transition from one state to another

Greek Letters Cheat Sheet (with LaTeX)

Symbol	LaTeX Code	Name	Pronunciation	Usage
α	<code>\alpha</code>	Alpha	AL-fuh	Variable, angle
β	<code>\beta</code>	Beta	BAY-tuh	Variable
γ	<code>\gamma</code>	Gamma	GAM-uh	Path, function
δ	<code>\delta</code>	Delta	DEL-tuh	Difference/change
ε	<code>\epsilon</code>	Epsilon	EP-suh-lon	Small quantity
θ	<code>\theta</code>	Theta	THAY-tuh	Angle
λ	<code>\lambda</code>	Lambda	LAM-duh	Functions, rates
μ	<code>\mu</code>	Mu	MYOO	Micro, mean

π	<code>\pi</code>	Pi	PIE	Ratio, constant
σ	<code>\sigma</code>	Sigma	SIG-muh	Summation, symbol
φ	<code>\phi</code>	Phi	FEE / FIE	Angle, function
ψ	<code>\psi</code>	Psi	SIGH	Wave function
ω	<code>\omega</code>	Omega	oh-MAY-guh	Angular frequency
Γ	<code>\Gamma</code>	Gamma (capital)	GAM-uh	Set, operator
Δ	<code>\Delta</code>	Delta (capital)	DEL-tuh	Change operator
Θ	<code>\Theta</code>	Theta (capital)	THAY-tuh	Threshold, bound
Λ	<code>\Lambda</code>	Lambda (capital)	LAM-duh	Type, logic
Π	<code>\Pi</code>	Pi (capital)	PIE	Product operator
Σ	<code>\Sigma</code>	Sigma (capital)	SIG-muh	Summation operator
Φ	<code>\Phi</code>	Phi (capital)	FEE / FIE	Symbol, field
Ψ	<code>\Psi</code>	Psi (capital)	SIGH	Symbol, function
Ω	<code>\Omega</code>	Omega (capital)	oh-MAY-guh	Limit, bound